

Organic Compounds Containing Nitrogen

Question1

Given below are two statements :

Statement-I : Benzenediazonium salt is prepared by the reaction of aniline with nitrous acid at $273 - 278$ K. It decomposes easily in the dry state.

Statement-II : Insertion of iodine into the benzene ring is difficult and hence iodobenzene is prepared through the reaction of benzenediazonium salt with KI.

In the light of the above statements, choose the most appropriate answer from the options given below:

NEET 2025

Options:

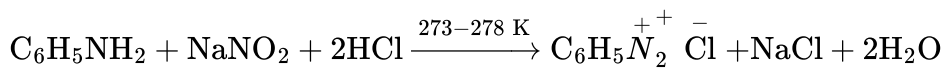
- A. Statement I is correct but Statement II is incorrect
- B. Statement I is incorrect but Statement II is correct
- C. Both Statement I and Statement II are correct
- D. Both Statement I and Statement II are incorrect

Answer: C

Solution:

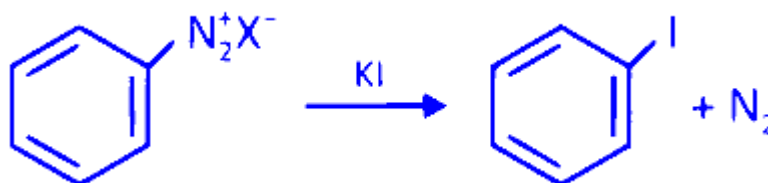
Benzene diazonium chloride is prepared by the reaction of aniline with nitrous acid at $273-278$ K .

Nitrous acid is produced in the reaction mixture by reaction of NaNO_2 with HCl .



Benzene diazonium chloride decomposes easily in the dry state

Iodobenzene is prepared by shaking benzene diazonium salt with KI because direct insertion of iodine into benzene ring is difficult



Question2

The correct order of decreasing basic strength of the given amines is:
Answer (1)Sol. Lower is the value of pK_b , higher is the basicity
 Also aliphatic amines are stronger bases than aromatic amines.
 pK_b : Benzenamine > N-Methylaniline > Ethanamine > N-Ethylethanamine
 Basic strength : N-Ethylethanamine > Ethanamine > N-Methylaniline > Benzenamine

NEET 2025

Options:

- A. N -ethylethanamine > ethanamine > N-methylaniline > benzenamine
- B. benzenamine > ethanamine > N-methylaniline > N-ethylethanamine
- C. N -methylaniline > benzenamine > ethanamine > N-ethylethanamine
- D. N-ethylethanamine > ethanamine > benzenamine > N-methylaniline

Answer: A

Solution:

Lower is the value of pK_b , higher is the basicity

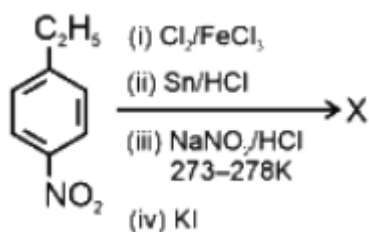
Also aliphatic amines are stronger bases than aromatic amines.

pK_b : Benzenamine > N-Methylaniline > Ethanamine > N-Ethylethanamine

Basic strength : N-Ethylethanamine > Ethanamine > N-Methylaniline > Benzenamine

Question3

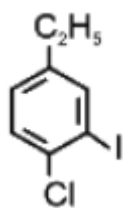
The major product X formed in the following reaction sequence is:



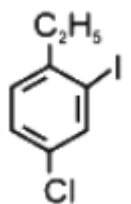
[NEET 2024 Re]

Options:

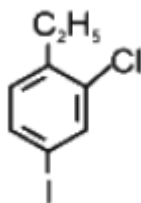
A.



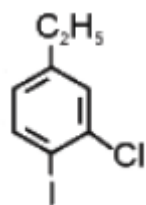
B.



C.

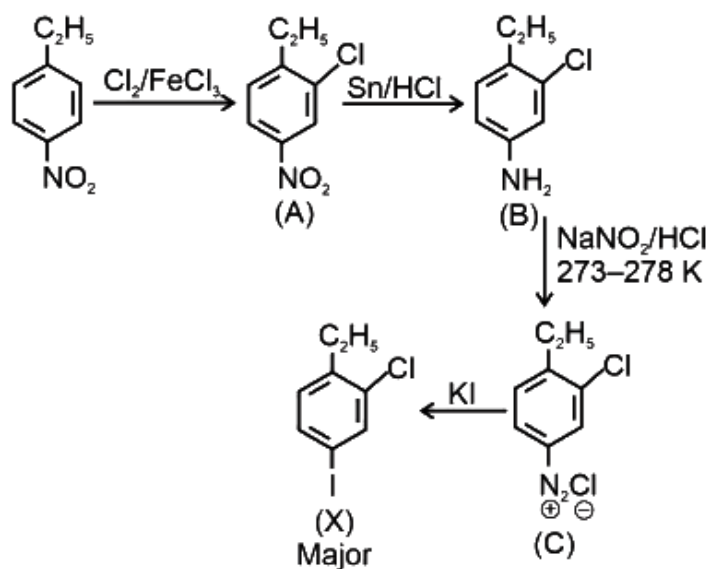


D.



Answer: C

Solution:



Question4

The compound that does not undergo Friedel-Crafts alkylation reaction but gives a positive carbylamine test is :

[NEET 2024 Re]

Options:

A.

Aniline

B.

Pyridine

C.

N -methylaniline

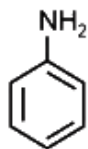
D.

Triethylamine

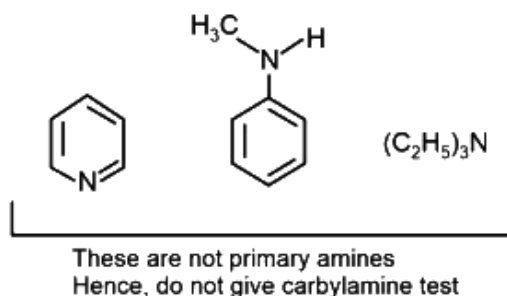
Answer: A

Solution:

For positive carbylamine test, there must be the presence of primary amine.



It will not show Friedal Crafts alkylation.
But gives positive carbylamine test



Question5

Given below are two statements:

Statement I : Aniline does not undergo Friedel-Crafts alkylation reaction.

Statement II : Aniline cannot be prepared through Gabriel synthesis.

In the light of the above statements, choose the correct answer from the options given below:

[NEET 2024]

Options:

A.

Both statement I and Statement II are true

B.

Both Statement I and Statement II are false

C.

Statement I is correct but Statement II is false

D.

Statement I is incorrect but Statement II is true

Answer: A

Solution:

- Aniline does not undergo Friedel-Crafts alkylation reaction due to salt formation with aluminium chloride, the Lewis acid, which is used as a catalyst.
- Aniline (aromatic primary amine) cannot be prepared by Gabriel phthalimide synthesis because aryl halides do not undergo nucleophilic substitution with anion formed by phthalimide.

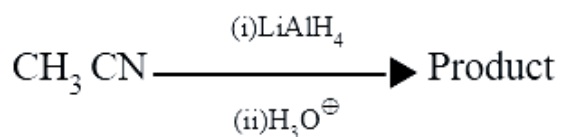
Question6

Which of the following reactions will NOT give primary amine as the product?

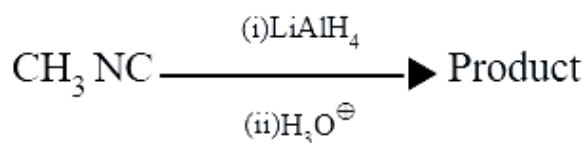
[NEET 2023]

Options:

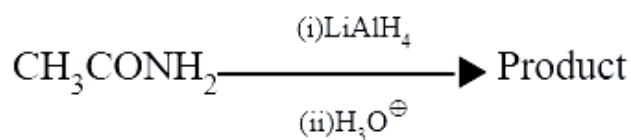
A.



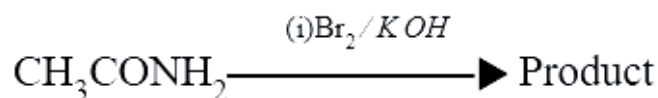
B.



C.



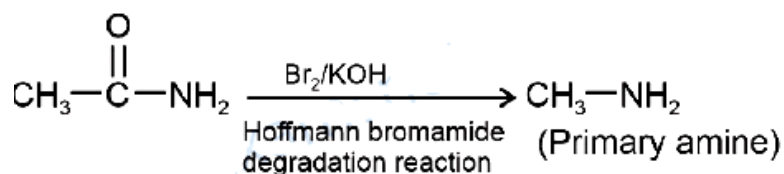
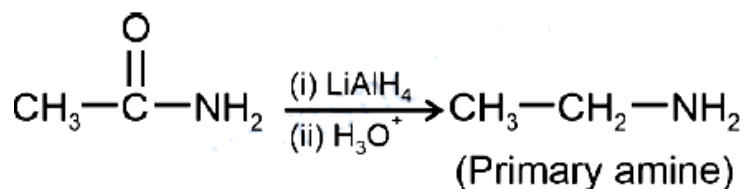
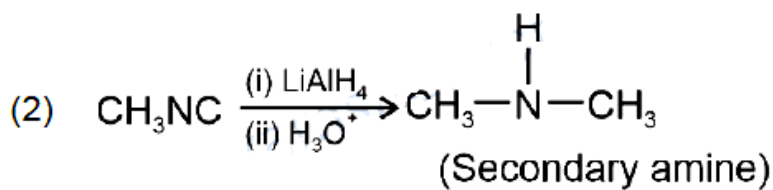
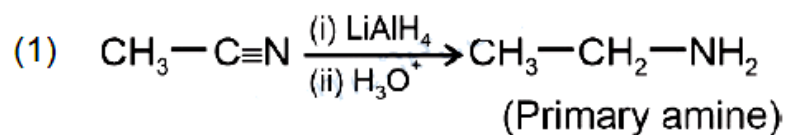
D.



Answer: B

Solution:

Solution:



Question7

Given below are two statements

Statement I:

The acidic strength of monosubstituted nitrophenol is higher than phenol because of electron withdrawing nitro group.

Statement II:

o-nitrophenol, m-nitrophenol and p-nitrophenol will have same acidic strength as they have one nitro group attached to the phenolic ring.

In the light of the above statements, choose the most appropriate answer from the options given below:

[NEET-2022]

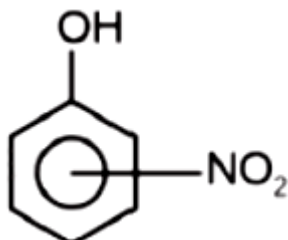
Options:

- A. Both Statement I and Statement II are correct.
- B. Both Statement I and Statement II are incorrect.
- C. Statement I is correct but Statement II is incorrect.
- D. Statement I is incorrect but Statement II is correct.

Answer: C

Solution:

Solution:



- Nitro group has electron withdrawing tendency. It can withdraw electrons both by $-I$ effect and $-R$ effect. Thus the acidic strength of monosubstituted nitrophenol is higher than phenol.
 - Nitro group present at *o*- and *p*-positions will have strong $-R$ effect while nitro group present at *m* position will influence only $-I$ effect hence acidity or *o/p* isomer will be more meta isomer.
-

Question8

Given below are two statements

Statement I:

Primary aliphatic amines react with HNO_2 to give unstable diazonium salts.

Statement II:

Primary aromatic amines react with HNO_2 to form diazonium salts

which are stable even above 300K .

In the light of the above statements, choose the most appropriate answer from the options given below

[NEET-2022]

Options:

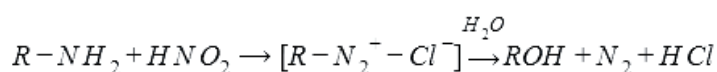
- A. Both Statement I and Statement II are correct.
- B. Both Statement I and Statement II are incorrect.
- C. Statement I is correct but Statement II is incorrect.
- D. Statement I is incorrect but Statement II is correct.

Answer: C

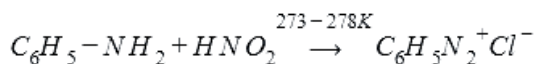
Solution:

Solution:

- Primary aliphatic amines react with HNO_2 and give unstable diazonium salt which turns into alcohol

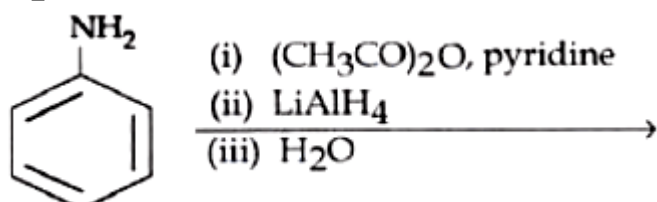


- Primary aromatic amines reacts with HNO_2 and give stable diazonium salt which are stable at 273 to 278K .



Question9

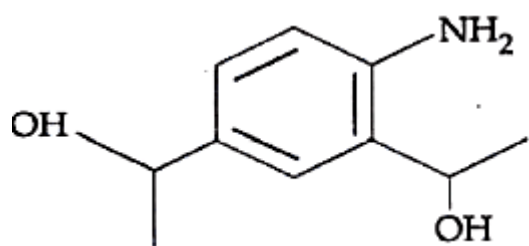
The product formed from the following reaction sequence is :



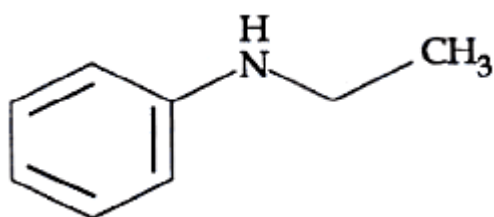
[NEET Re-2022]

Options:

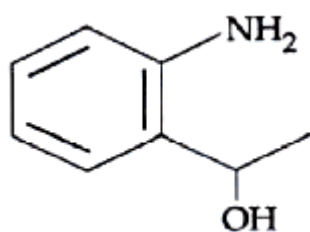
A.



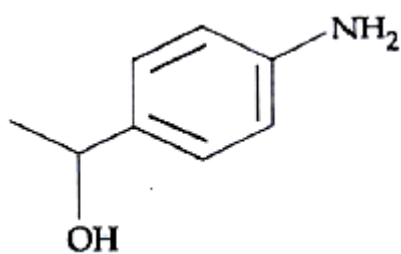
B.



C.



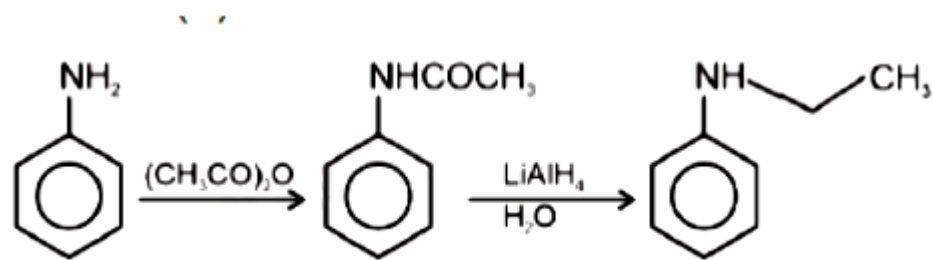
D.



Answer: B

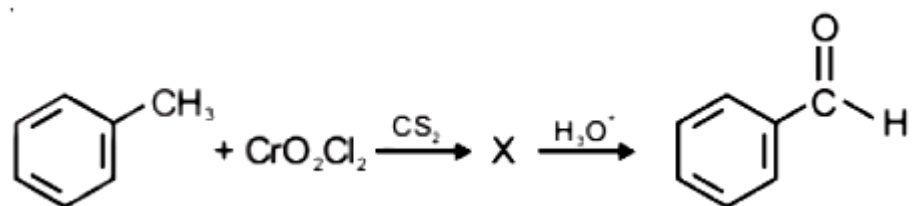
Solution:

Solution:



Question10

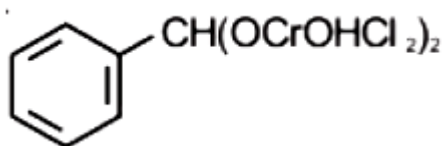
The intermediate compound 'X' in the following chemical reaction is:



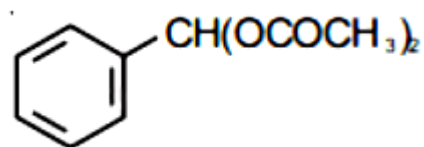
[NEET 2021]

Options:

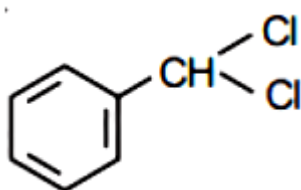
A.



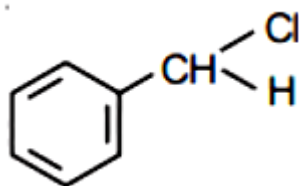
B.



C.



D.

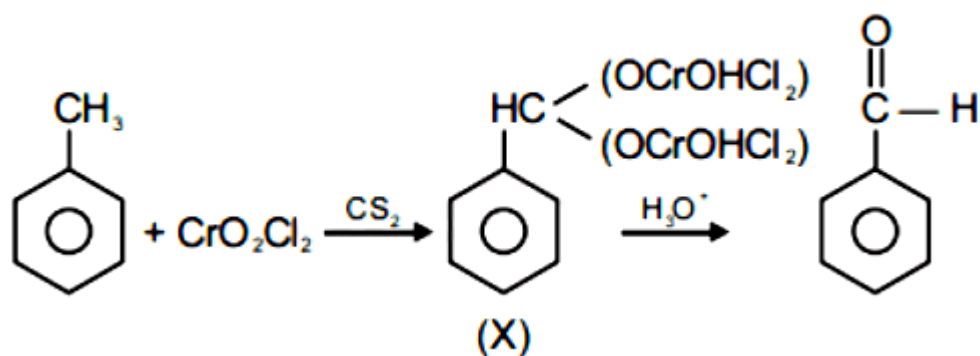


Answer: A

Solution:

Solution:

Etard's reaction

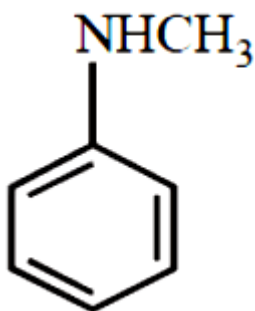


Question11

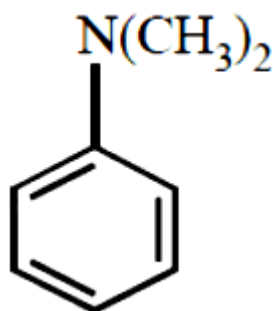
Which of the following amine will give the carbylamine test?
[2020]

Options:

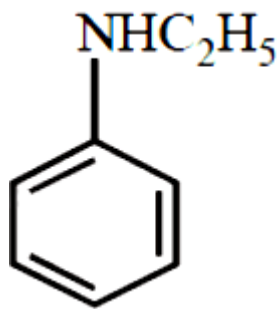
A.



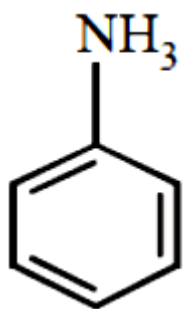
B.



C.



D.



Answer: D

Solution:

Solution:

Aliphatic and aromatic primary amines give carbylamine reaction. Since aniline is primary aromatic amine, it gives carbylamine test.

Question12

The correct order of the basic strength of methyl substituted amines in aqueous solution is (NEET 2019)

Options:

A. $\text{CH}_3\text{NH}_2 > (\text{CH}_3)_2\text{NH} > (\text{CH}_3)_3\text{N}$

B. $(\text{CH}_3)_2\text{NH} > \text{CH}_3\text{NH}_2 > (\text{CH}_3)_3\text{N}$

C. $(\text{CH}_3)_3\text{N} > \text{CH}_3\text{NH}_2 > (\text{CH}_3)_2\text{NH}$

D. $(\text{CH}_3)_3\text{N} > (\text{CH}_3)_2\text{NH} > \text{CH}_3\text{NH}_2$

Answer: B

Solution:

Solution:

The basicity of amines in aqueous solution depends on the stability of the ammonium cation or conjugate acid formed by accepting a proton from water which in turn depends on the +I -effect of alkyl group, extent of hydrogen bonding and steric factor.

All these factors are favourable for 2° amines.

Therefore, 2° amines are the strongest bases. If the alkyl group is small i.e., CH₃ then there is no steric hindrance to H-bonding.

Thus, the stability due to hydrogen bonding predominates over the stability due to +I -effect of CH₃ group and hence primary amine is a stronger base than 3° amine.

Hence, overall decreasing basic strength for methylamines in aqueous solution is

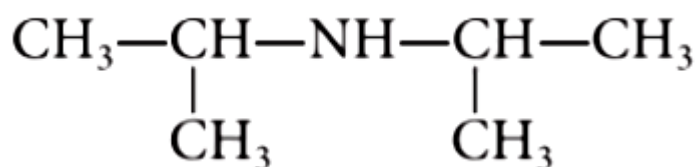


Question13

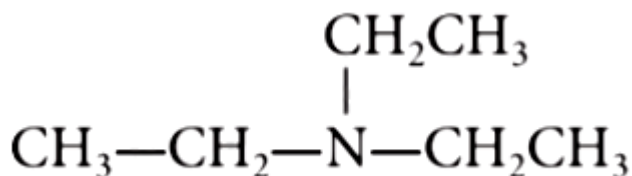
**The amine that reacts with Hinsberg's reagent to give an alkali insoluble product is
(Odisha NEET 2019)**

Options:

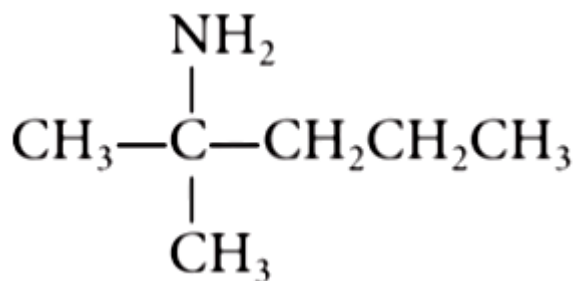
A.



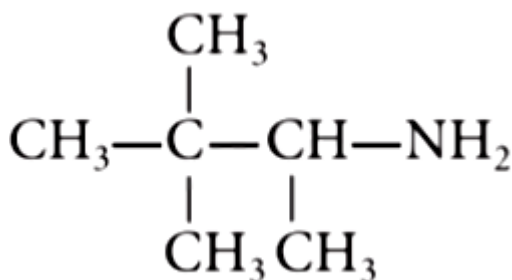
B.



C.



D.



Answer: A

Solution:

Solution:

Secondary amines on reaction with Hinsberg's reagent gives N, N -dialkylbenzene sulphonamide which does not contain any hydrogen atom attached to N -atom, it is not acidic and hence insoluble in alkali. Tertiary amines do not react with Hinsberg's reagent. Primary amines gives products which are soluble in alkali.

Question14

Nitration of aniline in strong acidic medium also gives m - nitroaniline because (NEET 2018)

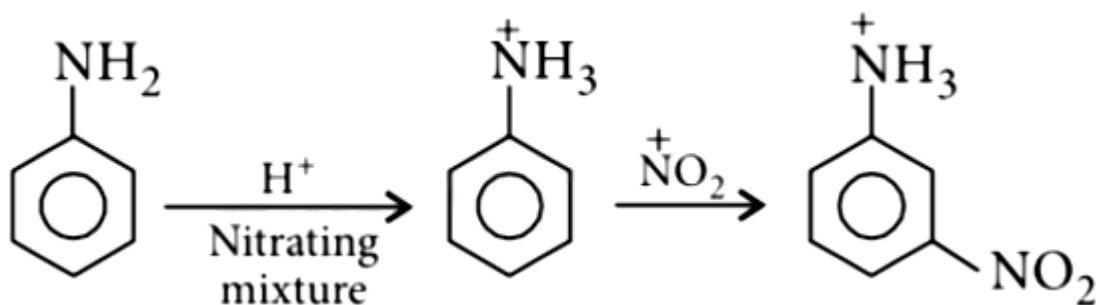
Options:

- A. inspite of substituents nitro group always goes to only m -position
- B. in electrophilic substitution reactions amino group is meta directive
- C. in absence of substituents nitro group always goes to m -position
- D. in acidic (strong) medium aniline is present as anilinium ion.

Answer: D

Solution:

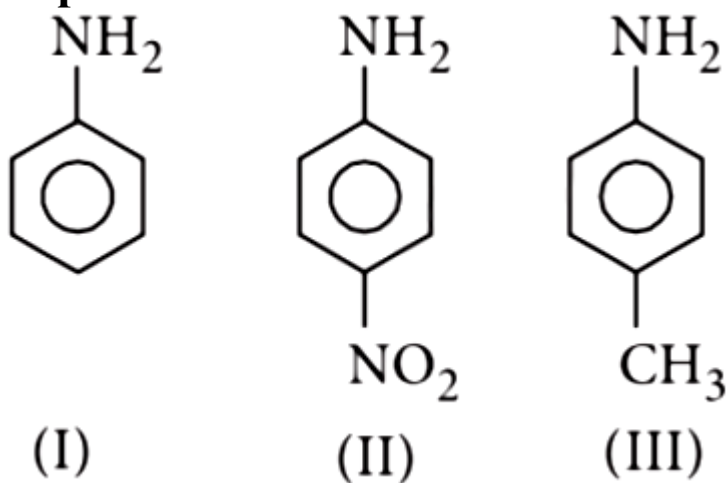
Solution:



The reason for formation of an unexpected amount of m -nitroaniline is that under strongly acidic condition of nitration, most of the aniline is converted into anilinium ion and since, NH_3^+ is a m - directing group, therefore, a large amount of m -nitroaniline is also obtained.

Question15

The correct increasing order of basic strength for the following compounds is



(NEET 2017)

Options:

- A. $\text{III} < \text{I} < \text{II}$
- B. $\text{III} < \text{II} < \text{I}$
- C. $\text{II} < \text{I} < \text{III}$
- D. $\text{II} < \text{III} < \text{I}$

Answer: C

Solution:

Solution:

Question16

Which of the following reactions is appropriate for converting acetamide to methanamine?
(NEET 2017)

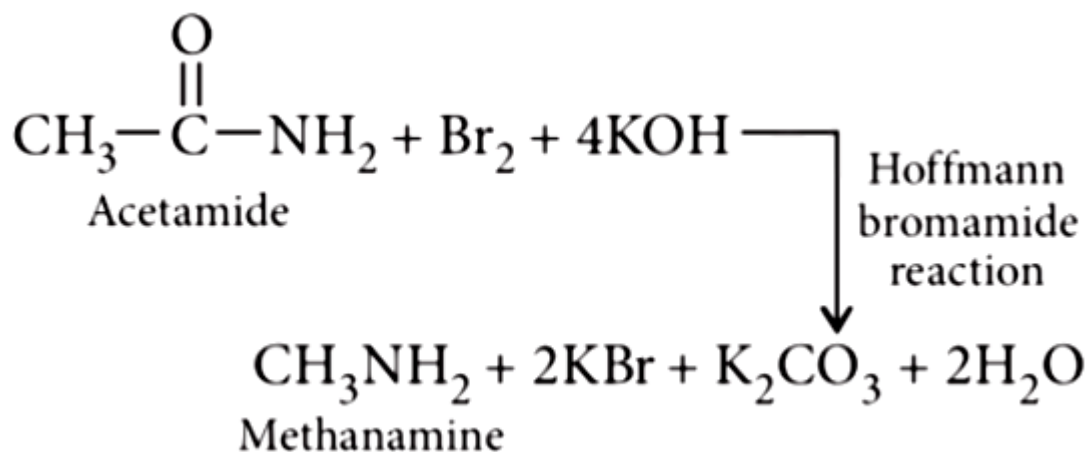
Options:

- A. Hoffmann hypobromamide reaction
- B. Stephen's reaction
- C. Gabriel phthalimide synthesis
- D. Carbylamine reaction

Answer: A

Solution:

Solution:



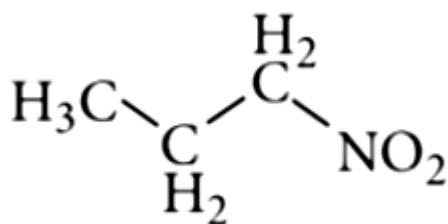
Question17

Which one of the following nitro-compounds does not react with nitrous acid?

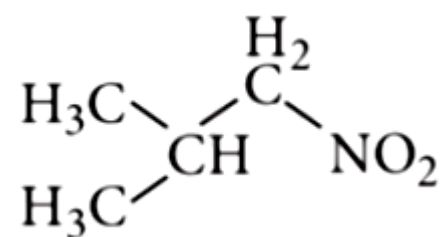
(NEET-II 2016)

Options:

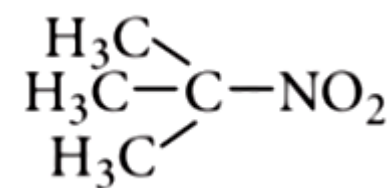
A.



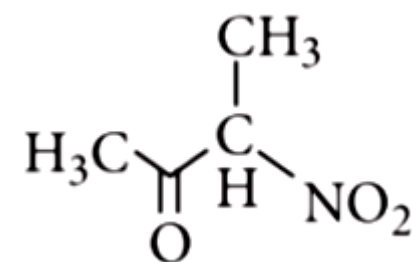
B.



C.



D.



Answer: C

Solution:

Solution:

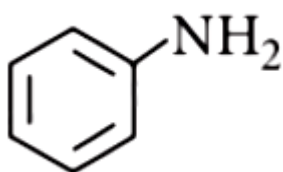
Tertiary nitroalkanes do not react with nitrous acid as they do not contain α -hydrogen atom.

Question18

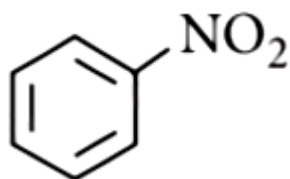
A given nitrogen-containing aromatic compound 'A' reacts with Sn/HCl , followed by HNO_2 to give an unstable compound 'B'. 'B', on treatment with phenol, forms a beautiful coloured compound 'C' with the molecular formula $\text{C}_{12}\text{H}_{10}\text{N}_2\text{O}$. The structure of compound 'A' is
(NEET-II 2016)

Options:

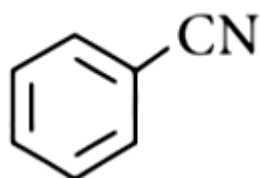
A.



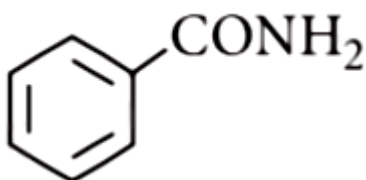
B.



C.



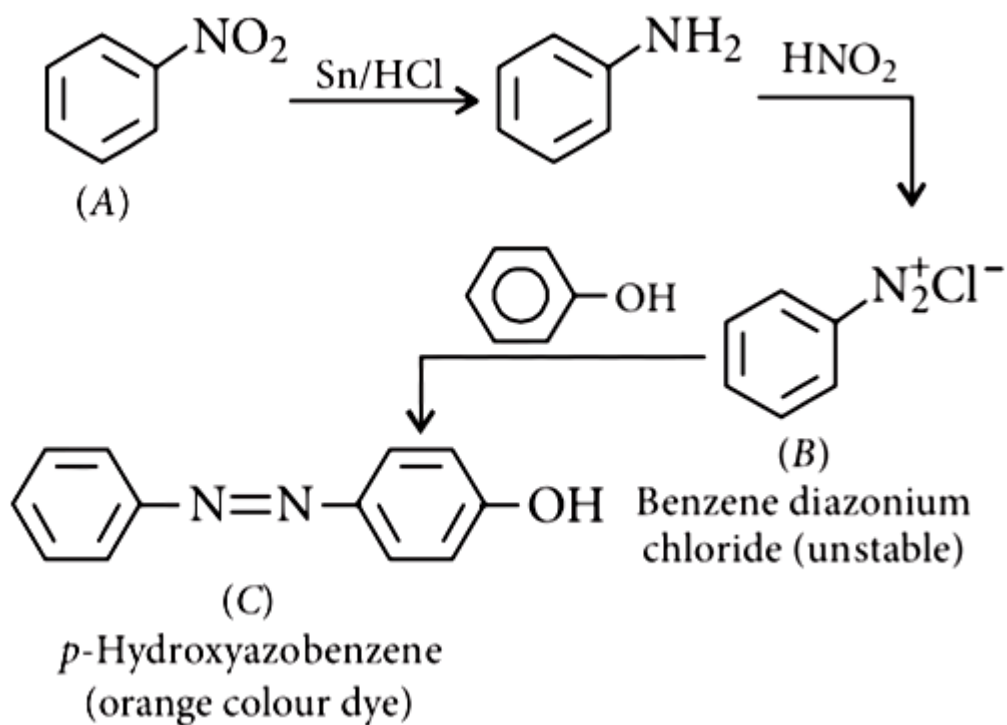
D.



Answer: B

Solution:

Solution:



Question19

The correct statement regarding the basicity of arylamines is (NEET-I 2016)

Options:

- A. arylamines are generally more basic than alkylamines because of aryl group
- B. arylamines are generally more basic than alkylamines, because the nitrogen atom in arylamines is sp - hybridised
- C. arylamines are generally less basic than alkylamines because the nitrogen lone-pair electrons are delocalised by interaction with the aromatic ring π -electron system
- D. arylamines are generally more basic than alkylamines because the nitrogen lone-pair electrons are not delocalised by interaction with the aromatic ring π -electron system.

Answer: C

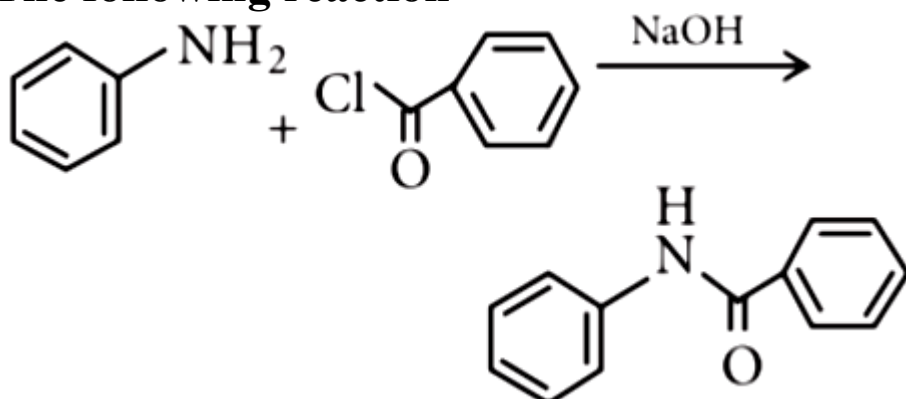
Solution:

Solution:

In arylamines, lone pair of electrons on nitrogen atom is delocalised over the benzene ring, thus, not available for donation. So, arylamines are less basic than alkylamines.

Question20

The following reaction



is known by the name
(2015)

Options:

- A. Perkin's reaction
- B. Acetylation reaction
- C. Schotten-Baumann reaction
- D. Friedel-Crafts reaction.

Answer: C

Solution:

Solution:

Benzoylation of compounds containing an active hydrogen atom such as alcohols, phenols and amines with benzoyl chloride in the presence of dilute aq. NaOH solution is called Schotten-Baumann reaction.

Question21

Method by which aniline cannot be prepared is
(2015)

Options:

- A. degradation of benzamide with bromine in alkaline solution
- B. reduction of nitrobenzene with H_2/Pd in ethanol
- C. potassium salt of phthalimide treated with chlorobenzene followed by hydrolysis with aqueous NaOH solution
- D. hydrolysis of phenylisocyanide with acidic solution.

Answer: C

Solution:

Solution:

Aniline cannot be prepared by this method because aryl halides do not undergo nucleophilic substitution reaction with potassium phthalimide under mild conditions.

Question22

**The electrolytic reduction of nitrobenzene in strongly acidic medium produces
(2015 Cancelled)**

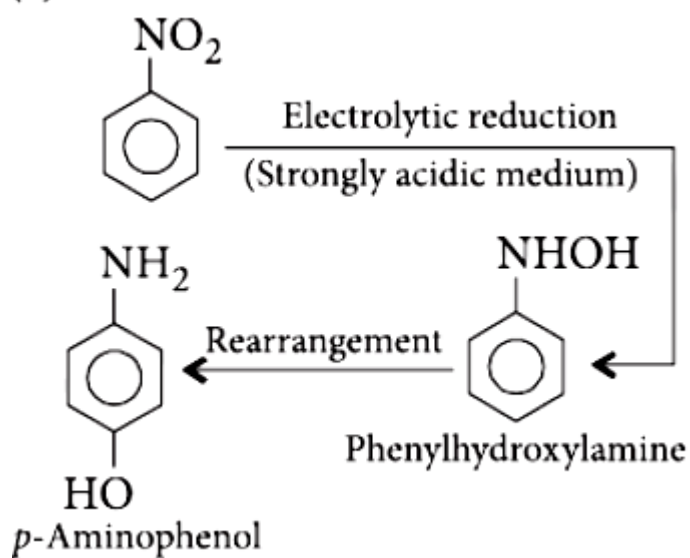
Options:

- A. azobenzene
- B. aniline
- C. p-aminophenol
- D. azoxybenzene.

Answer: C

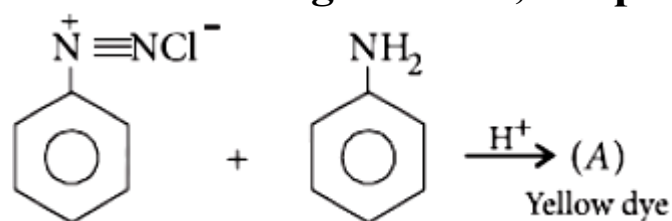
Solution:

Solution:



Question23

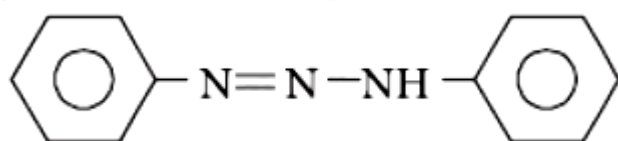
In the following reaction, the product (A)



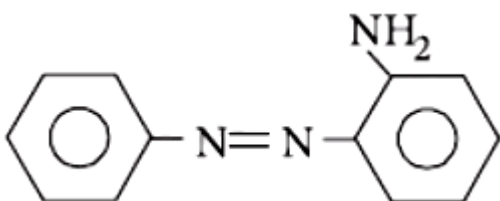
(2014)

Options:

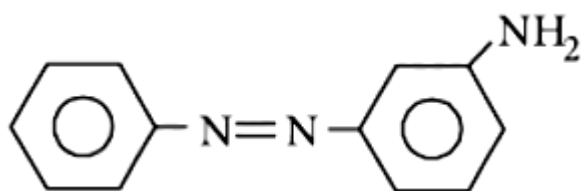
A.



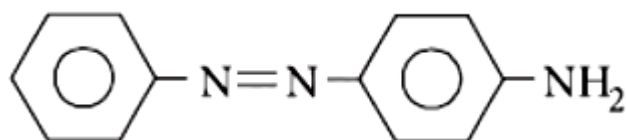
B.



C.



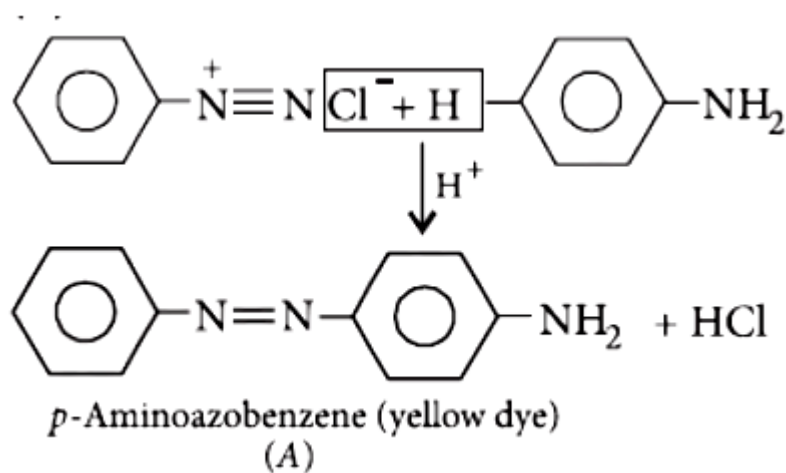
D.



Answer: D

Solution:

Solution:



Question24

Which of the following will be most stable diazonium salt RN_2^+X^- ?
(2014)

Options:





Answer: B

Solution:

Solution:

Aromatic diazonium salts are more stable due to dispersal of the positive charge in benzene ring.

Question25

**Nitrobenzene on reaction with conc. $\text{HNO}_3/\text{H}_2\text{SO}_4$ at $80-100^\circ\text{C}$ forms which one of the following products?
(2013 NEET)**

Options:

A. 1, 4-Dinitrobenzene

B. 1, 2, 4-Trinitrobenzene

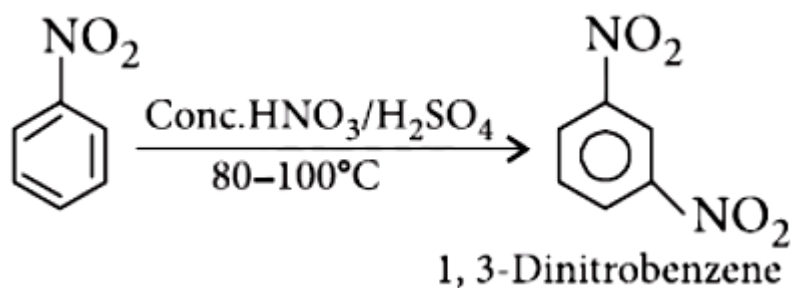
C. 1, 2-Dinitrobenzene

D. 1, 3-Dinitrobenzene

Answer: D

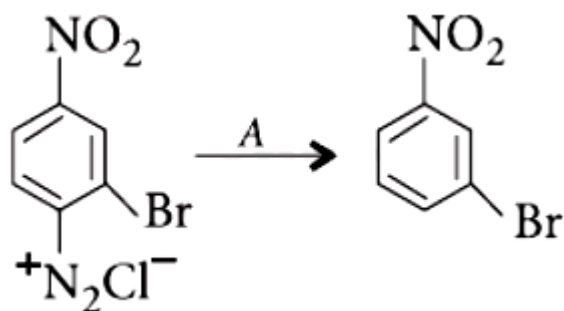
Solution:

Solution:



Question26

In the reaction



(2013 NEET)

Options:

A. H_3PO_2 and H_2O

B. $\text{H}^+/\text{H}_2\text{O}$

C. $\text{HgSO}_4/\text{H}_2\text{SO}_4$

D. CuCl_2

Answer: A

Solution:

Solution:

H_3PO_2 and H_2O reduces the $-\text{N}_2^+\text{Cl}^-$ to $-\text{H}$

Question27

On hydrolysis of a "compound", two compounds are obtained. One of which on treatment with sodium nitrite and hydrochloric acid gives a product which does not respond to iodoform test. The second one reduces Tollens' reagent and Fehling's solution. The "compound" is
(Karnataka NEET 2013)

Options:

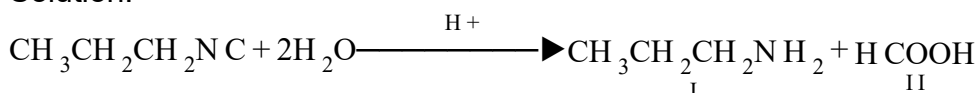
A. $\text{CH}_3\text{CH}_2\text{CH}_2\text{NC}$



Answer: A

Solution:

Solution:



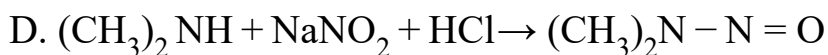
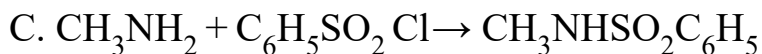
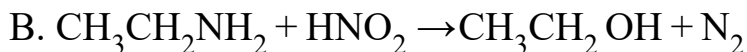
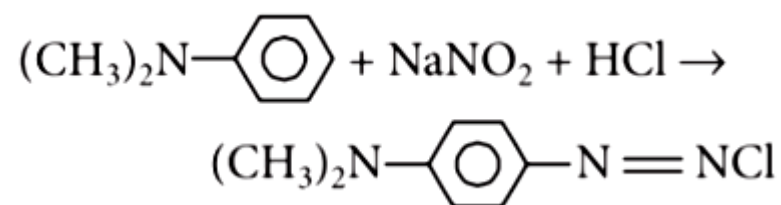
II (HCOOH) reduces Tollens'reagent and Fehling's solution.

Question28

**Some reactions of amines are given. Which one is not correct?
(Karnataka NEET 2013)**

Options:

A.

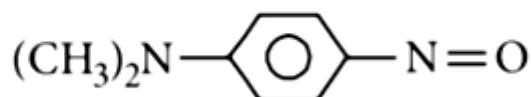
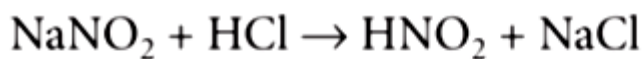


Answer: A

Solution:

Solution:

Aromatic tertiary amines undergo electrophilic substitution with nitrosonium ion at p -position of the phenyl ring to form green coloured p -nitrosoamines.



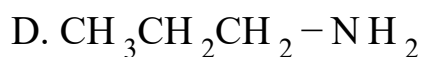
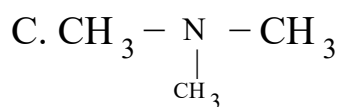
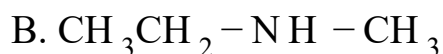
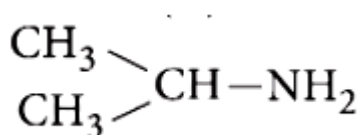
p-Nitro-*N*, *N*-dimethyl
aniline (green)

Question29

An organic compound ($\text{C}_3\text{H}_9\text{N}$)(A) when treated with nitrous acid, gave an alcohol and N_2 gas was evolved. (A) on warming with CHCl_3 and caustic potash gave (C) which on reduction isopropylmethanamine. Predict the structure of (A). (2012 Mains)

Options:

A.

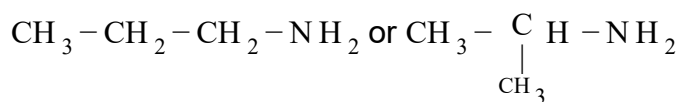


Answer: A

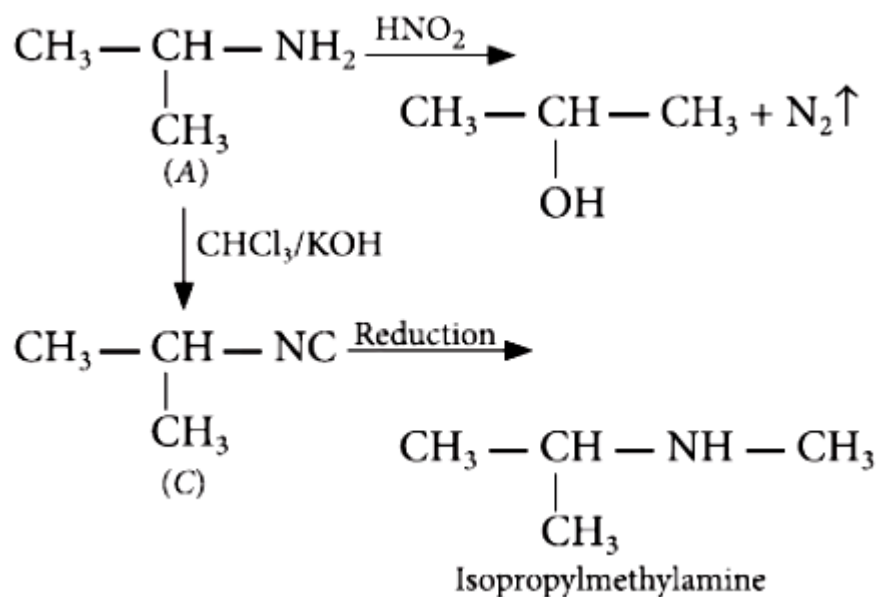
Solution:

Solution:

As A gives alcohol on treatment with nitrous acid thus it should be primary amine. $\text{C}_3\text{H}_9\text{N}$ has two possible structures with $-\text{NH}_2$ group

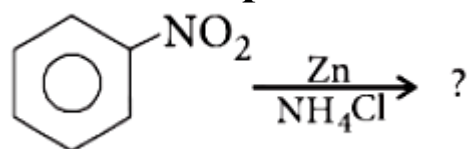


As it gives isopropyl methylamine thus it should be isopropyl amine not n-propyl amine.



Question30

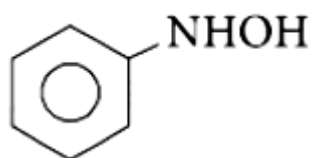
What is the product obtained in the following reaction ?



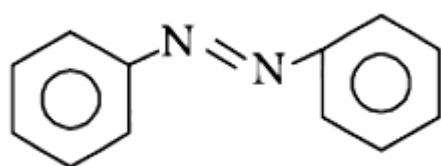
(2011)

Options:

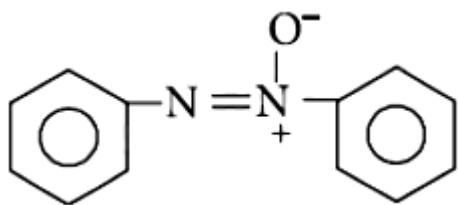
A.



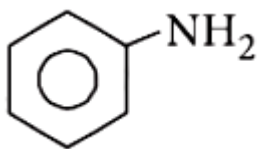
B.



C.



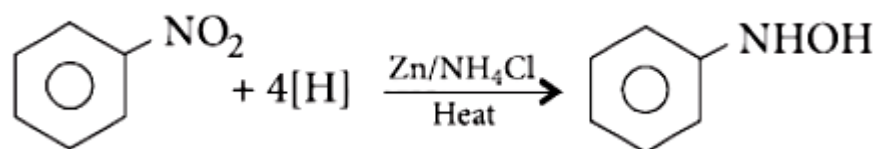
D.



Answer: A

Solution:

Solution:

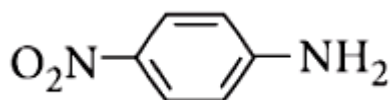


Question31

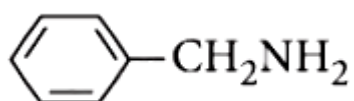
**Which of the following compounds is most basic?
(2011 Mains)**

Options:

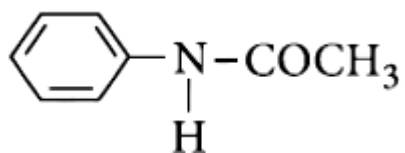
A.



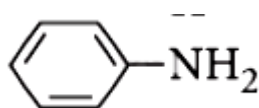
B.



C.



D.



Answer: B

Solution:

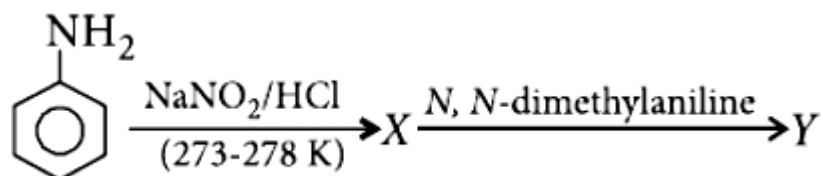
Solution:

In benzylamine the electron pair present on the nitrogen is not delocalised with the benzene ring.

Question32

Aniline in a set of the following reactions yielded a coloured product Y .

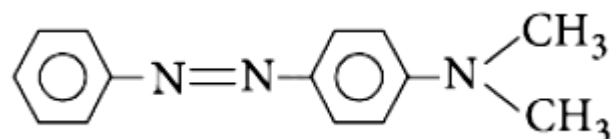
The structure of ' Y ' would be



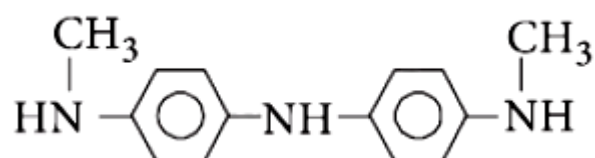
(2008,2010)

Options:

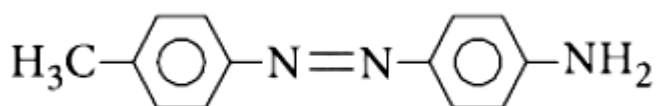
A.



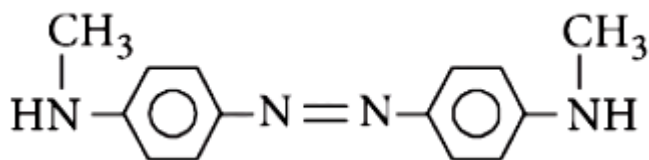
B.



C.



D.



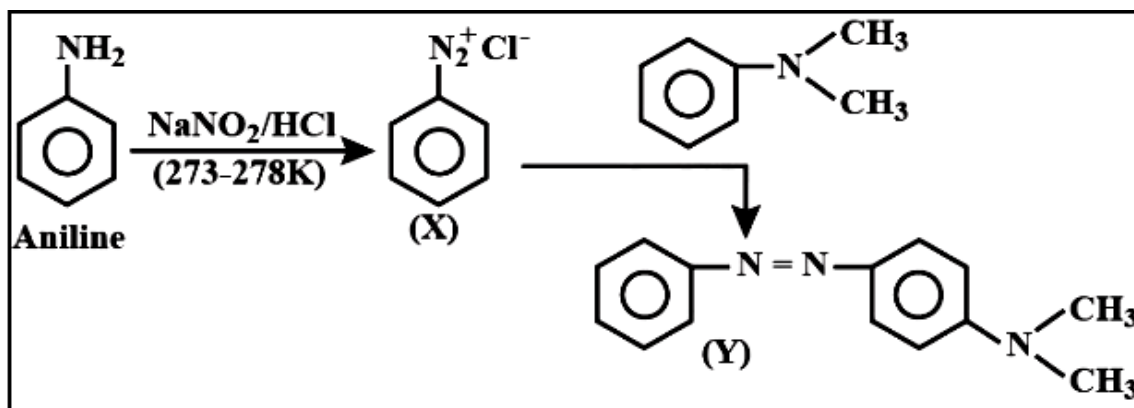
Answer: A

Solution:

Solution:

The first step is diazotization reaction which form [X] which on further reaction with N , N – dimethylaniline gives product Y

It attached to the para position as (N – (CH₃)₂) group is ortho-para activating group. Electrophilic addition takes place on para position.



Question33

**Which of the following statements about primary amines is false?
(2010)**

Options:

- A. Alkyl amines are stronger bases than aryl amines
- B. Alkyl amines react with nitrous acid to produce alcohols.
- C. Aryl amines react with nitrous acid to produce phenols

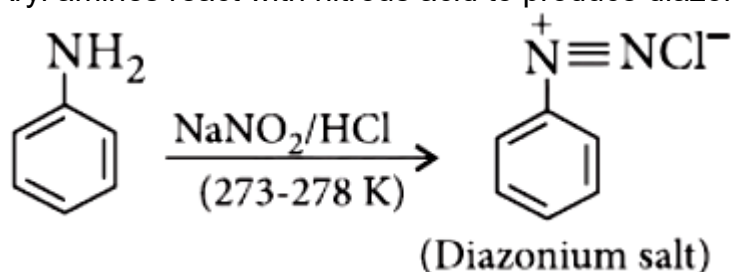
D. Alkyl amines are stronger bases than ammonia.

Answer: C

Solution:

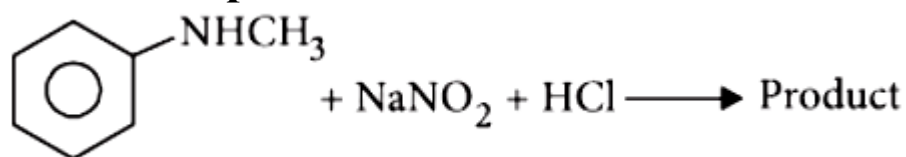
Solution:

Aryl amines react with nitrous acid to produce diazonium salts.



Question34

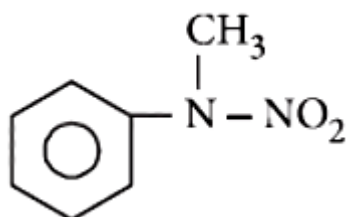
Predict the product



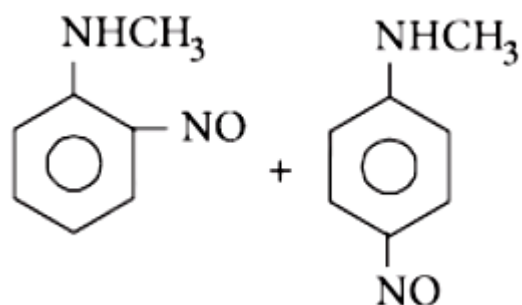
(2009)

Options:

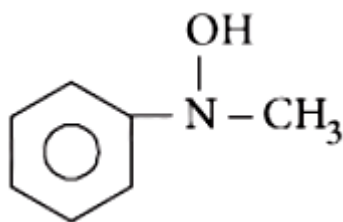
A.



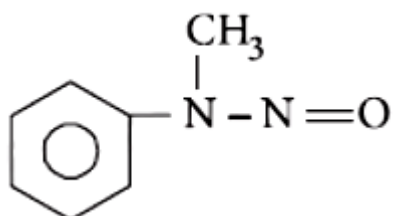
B.



C.



D.

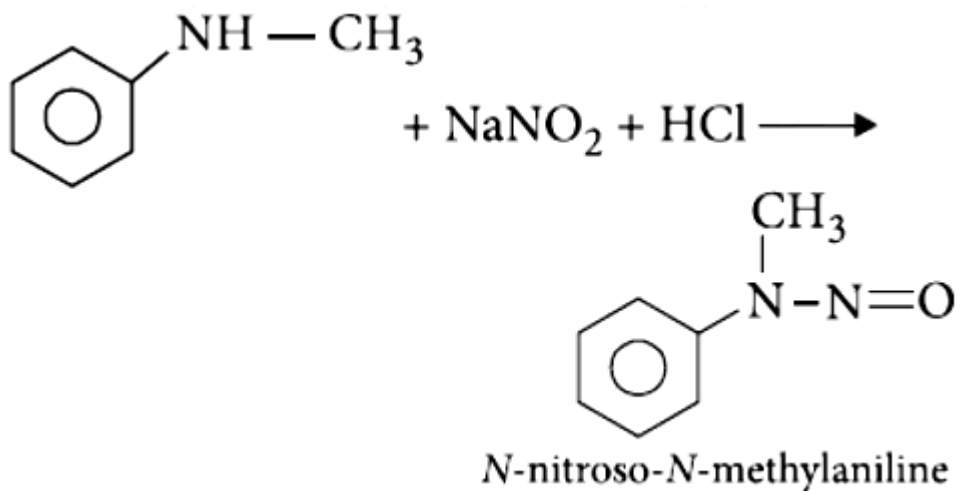


Answer: D

Solution:

Solution:

2° aliphatic and aromatic amines react with nitrous acid to form N-nitrosoamine.



Question35

Nitrobenzene can be prepared from benzene by using a mixture of conc.HNO₃ and conc.H₂SO₄.In the mixture ,nitric acid acts as a/an (2009)

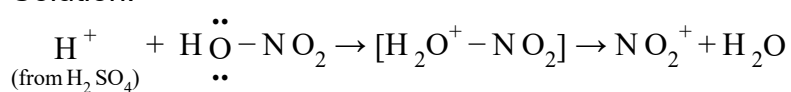
Options:

- A. acid
- B. base
- C. catalyst
- D. reducing agent

Answer: B

Solution:

Solution:



Question 36

**Which one of the following on reduction with lithium aluminium hydride yields a secondary amine?
(2007)**

©

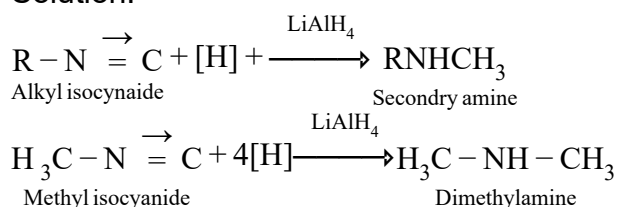
Options:

- A. Methyl isocyanide
- B. Acetamide
- C. Methyl cyanide
- D. Nitroethane.

Answer: A

Solution:

Solution:



Question37

**Which of the following is more basic than aniline?
(2006)**

©

Options:

- A. Benzylamine
- B. Diphenylamine
- C. Triphenylamine
- D. p-Nitroaniline

Answer: A

Solution:

Solution:

Any group which when present on benzene ring has electron withdrawing ($-\text{NO}_2$, $-\text{CN}$, $-\text{SO}_3\text{H}$, $-\text{COOH}$, $-\text{Cl}$, $-\text{C}_6\text{H}_5$, etc) group decreases basicity of aniline, e.g. aniline is more basic than nitroaniline. Lone pair electrons are more delocalised in diphenylamine and triphenylamine, thus these are less basic than aniline. In benzylamine the electron pair present on nitrogen is not delocalised with the benzene ring hence it is more basic than aniline.

Question38

**Electrolytic reduction of nitrobenzene in weakly acidic medium gives
(2005)**

©

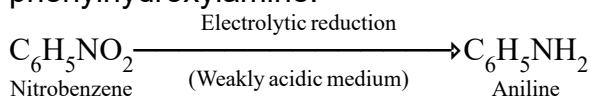
Options:

- A. N -phenylhydroxylamine
- B. nitrosobenzene

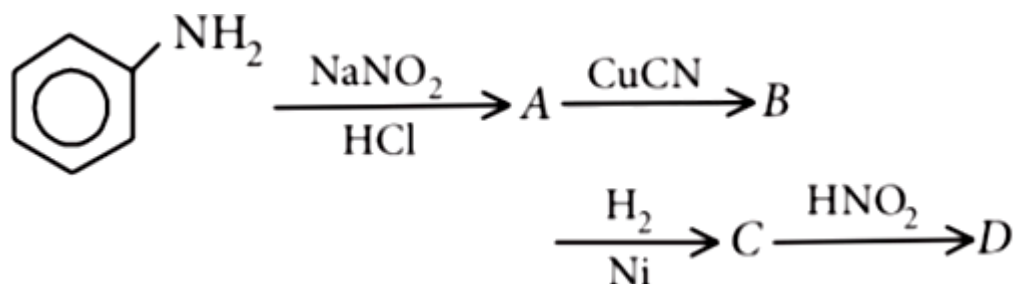
D. p -hydroxyaniline.

Solution:

Electrolytic reduction of nitrobenzene in weakly acidic medium gives aniline but in strongly acidic medium it gives p -aminophenol through the acid-catalysed rearrangement of the initially formed phenylhydroxylamine.



Aniline in a set of reactions yielded a product D.



Options:

A. C₆H₅NHOH

B. $\text{C}_6\text{H}_5\text{NHCH}_2\text{CH}_3$

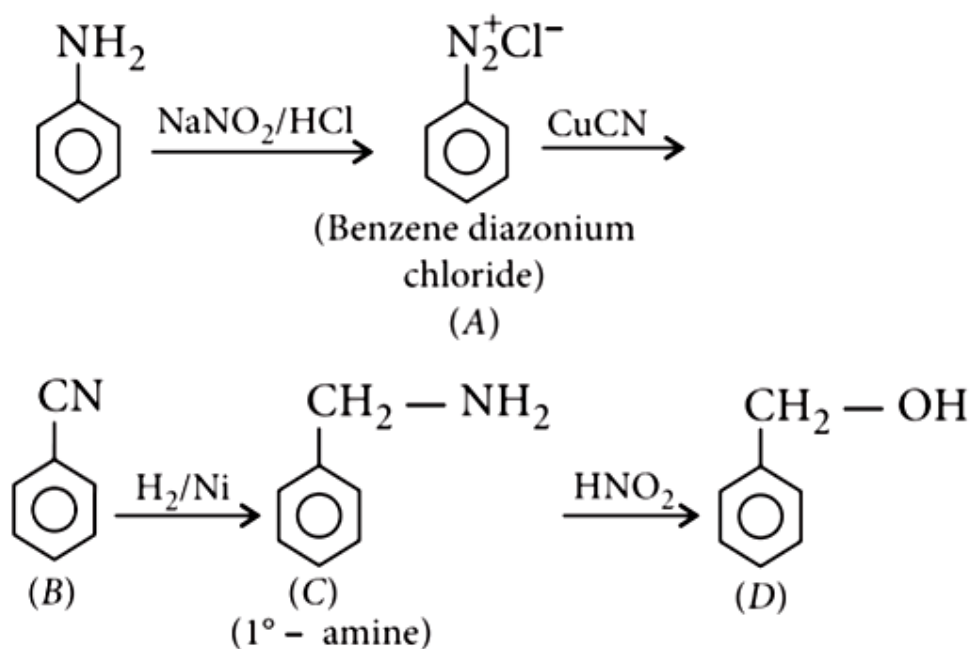
C. $\text{C}_6\text{H}_5\text{CH}_2\text{NH}_2$

D. $\text{C}_6\text{H}_5\text{CH}_2\text{OH}$

Answer: D

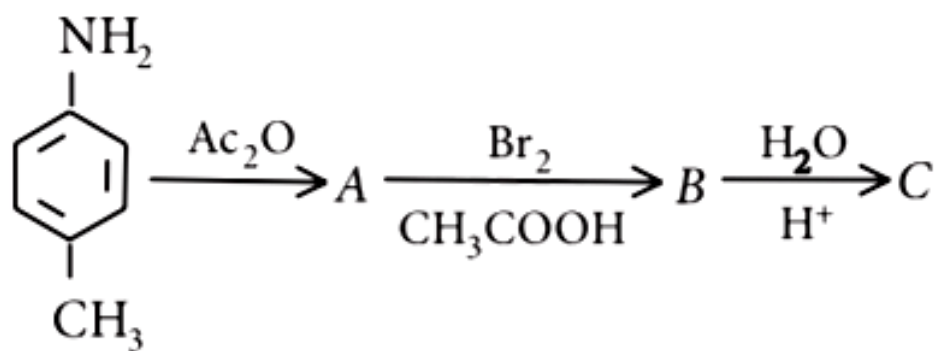
Solution:

Solution:



Question40

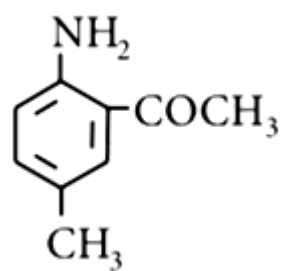
The final product C, obtained in the given reaction would be :



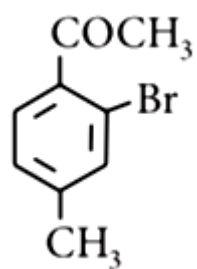
(2003)

Options:

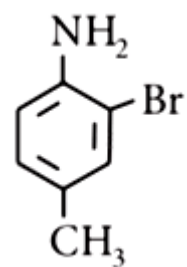
A.



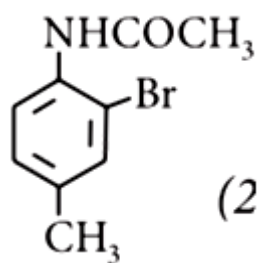
B.



C.



D.

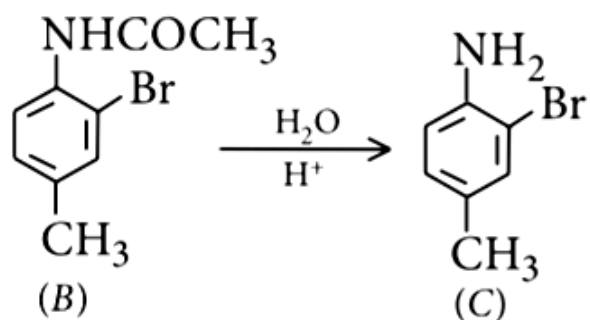
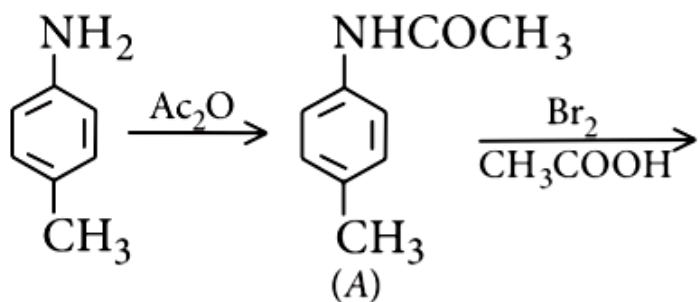


(2

Answer: C

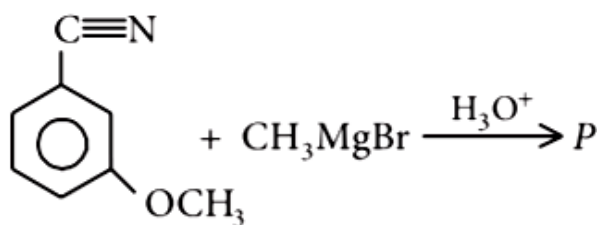
Solution:

Solution:



Question41

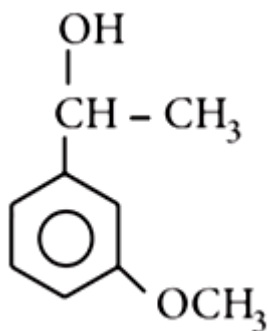
Product 'P' in the above reaction is



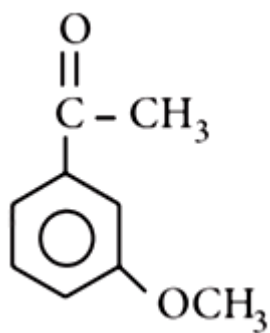
(2002)

Options:

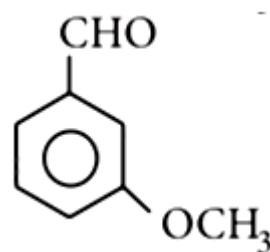
A.



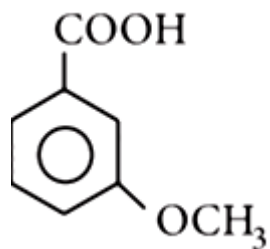
B.



C.



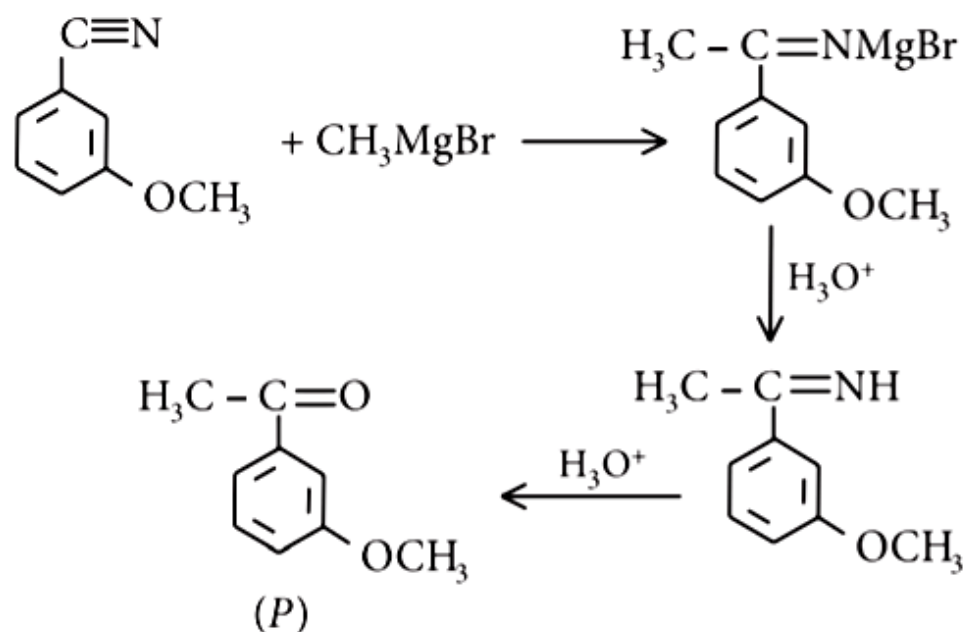
D.



Answer: B

Solution:

Solution:



Question42

Intermediates formed during reaction of RCONH_2 with Br_2 and KOH are (2001)

Options:

- A. R CONHBr and R NCO
- B. R NHCOBr and R NCO
- C. $\text{R NH}-\text{Br}$ and RCONHBr
- D. RCONBr_2

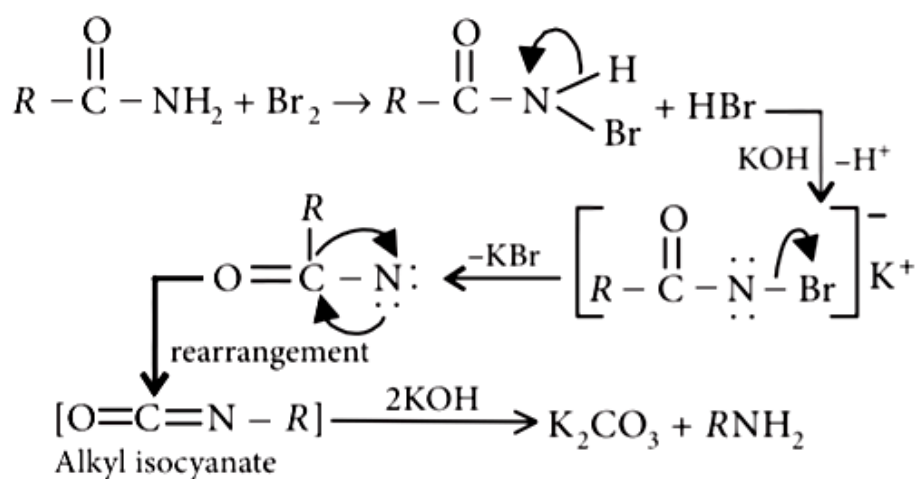
Answer: A

Solution:

Solution:

The reaction, $\text{RCONH}_2 + \text{Br}_2 + \text{KOH} \rightarrow \text{RNH}_2$ is known as Hoffmann-bromamide reaction.

The mechanism of the reaction is



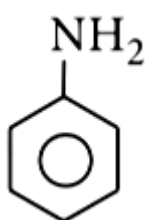
This reaction is used to descent the series, i.e., for preparing a lower homologue from a higher one.

Question43

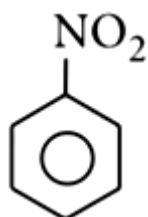
$\text{A} \xrightarrow{\text{reduction}} \text{B} \xrightarrow{\text{CHCl}_3/\text{KOH}} \text{C} \xrightarrow{\text{reduction}} \text{N-methylaniline},$
 then A is
 (2000)

Options:

A.

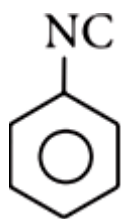


B.



C. CH_3NH_2

D.



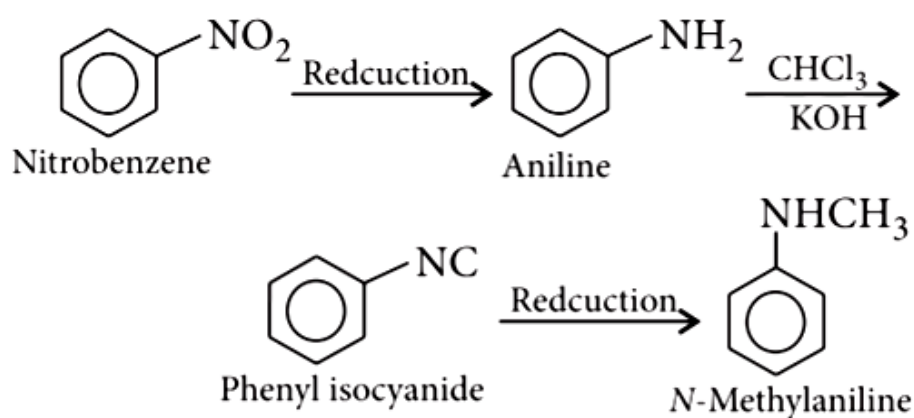
Answer: B

Solution:

Solution:

'C' must be an isocyanide and obtained from a 1° amine by carbylamine reaction ($\text{CHCl}_3 + \text{KOH}$).

Further 1° amine must be obtained by reduction of nitro compounds. So 'A' is nitrobenzene.



Question44

Amides may be converted into amines by a reaction named after (1999)

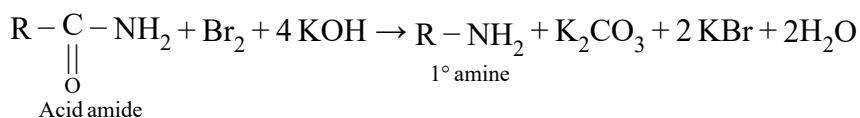
Options:

- A. Hoffmann
- B. Claisen
- C. Perkin
- D. Kekule.

Answer: A

Solution:

Solution:



This reaction is called Hoffmann bromamide reaction.

Question45

**Phenyl isocyanides are prepared by which of the following reaction?
(1999)**

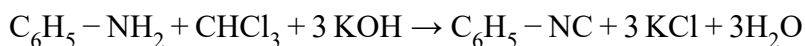
Options:

- A. Reimer-Tiemann reaction
- B. Carbylamine reaction
- C. Rosenmund's reaction
- D. Wurtz reaction

Answer: B

Solution:

Solution:



The above reaction is called as carbylamine reaction, which is a specific reaction of 1° -amine.

Question46

**Aniline is reacted with bromine water and the resulting product is treated with an aqueous solution of sodium nitrite in presence of dilute hydrochloric acid. The compound so formed is converted into a tetrafluoroborate which is subsequently heated to dry. The final product is
(1998)**

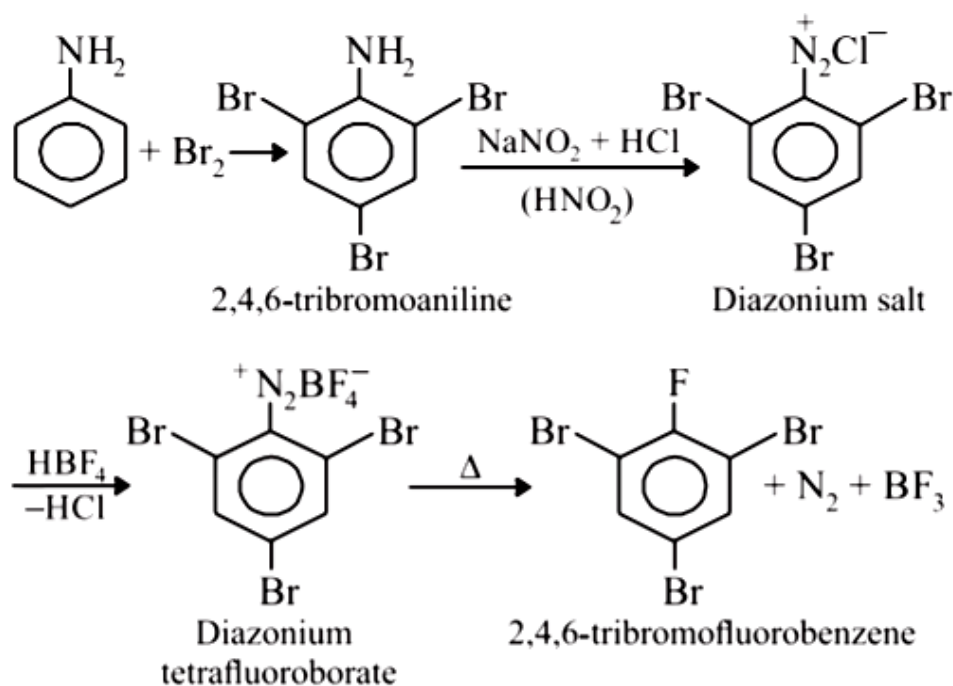
Options:

- A. p -bromoaniline
- B. p -bromofluorobenzene
- C. 1,3,5 -tribromobenzene
- D. 2,4,6 -tribromofluorobenzene.

Answer: D

Solution:

Solution:



Question47

The compound obtained by heating a mixture of ethylamine and chloroform with ethanolic potassium hydroxide (KOH) is (1997)

Options:

- A. an amide
- B. an amide and nitro compound

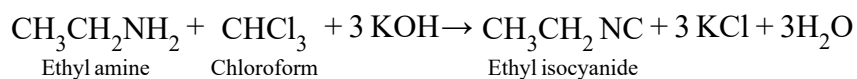
C. an ethyl isocyanide

D. an alkyl halide.

Answer: C

Solution:

Solution:

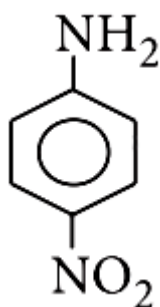


Question48

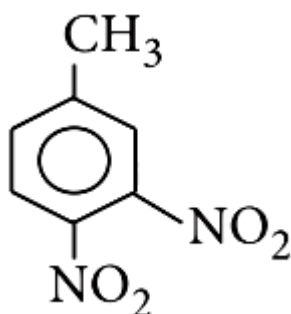
**An aniline on nitration gives
(1996)**

Options:

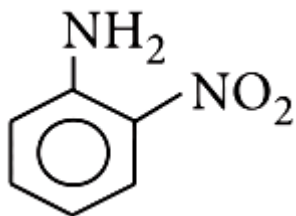
A.



B.



C.

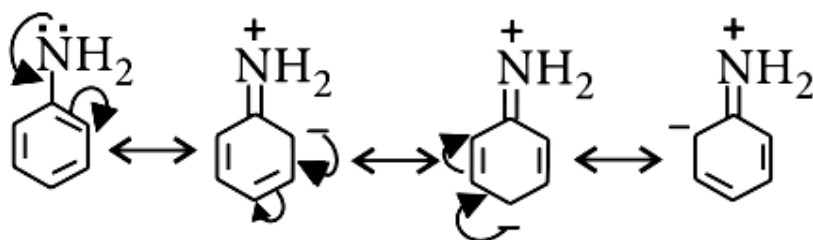


D. both (a) and (c)

Answer: D

Solution:

Solution:



As, NO_2^+ electrophile can attack both ortho and para positions, therefore both (a) and (c) product will be obtained.

Question49

**Which product is formed, when acetonitrile is hydrolysed partially with cold concentrated HCl?
(1995)**

©

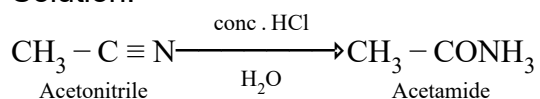
Options:

- A. Methyl cyanide
- B. Acetic anhydride
- C. Acetic acid
- D. Acetamide

Answer: D

Solution:

Solution:



Question50

The action of nitrous acid on an aliphatic primary amine gives (1994)

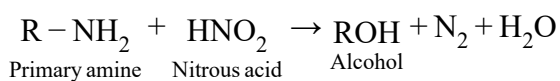
Options:

- A. secondary amine
- B. nitro alkane
- C. alcohol
- D. alkyl nitrite.

Answer: C

Solution:

Solution:



Question51

Which one of the following order is wrong, with respect to the property indicated? (1994)

Options:

- A. Benzoic acid > phenol > cyclohexanol (acid strength)
- B. Aniline > cyclohexylamine > benzamide (basic strength)

C. Formic acid > acetic acid > propanoic acid (acid strength)

D. Fluoroacetic acid > chloroacetic acid > bromoacetic acid (acid strength)

Answer: B

Solution:

Solution:

Basic strength decreases as, cyclohexylamine > aniline > benzamide.

Lesser basicity in aniline and benzamide is due to participation of lone pair of electron of - NH₂ group in resonance.

Question52

For carbylamine reaction, we need hot alcoholic KOH and (1992)

Options:

A. any primary amine and chloroform

B. chloroform and silver powder

C. a primary amine and an alkyl halide

D. a monoalkylamine and trichloromethane.

Answer: A

Solution:

Solution:

In carbylamine reaction, primary amines on heating with chloroform in presence of alcoholic KOH form isocyanides (or carbylamines). It is used to distinguish 1° amines from 2° and 3° amines.



Question53

Indicate which nitrogen compound amongst the following would undergo Hoffmann reaction (i.e., reaction with Br_2 and strong KOH) to furnish the primary amine ($\text{R} - \text{NH}_2$)? (1989)

Options:



Answer: C

Solution:

Solution:

The amide ($-\text{CONH}_2$) group is converted into primary amino group ($-\text{NH}_2$) by Hoffmann's bromamide reaction.

